THE C# ARRAYLIST

WHAT IS AN ARRAYLIST?

An ArrayList is a dynamic collection in C# that can store elements of different data types and can automatically resize.

Key Characteristics:

- Dynamic size (no fixed length)
- Stores objects (object type)
- Allows duplicate values

```
ArrayList list = new ArrayList();
```

CREATING AN ARRAYLIST

To use an ArrayList, you must include the required namespace.

```
using System.Collections;
```

Declaration of ArrayList

```
ArrayList list = new ArrayList();
```

With Initial values

```
ArrayList items = new ArrayList() { 1, "Apple", 3.5 };
```

ARRAYLIST CLASS

The ArrayList class provides built-in methods to manage data.

Common Properties and Methods:

- Add() – adds an element
- Remove() – removes a specific element
- RemoveAt() – removes by index
- Count – number of elements
- Clear() – removes all elements

ARRAYLIST CLASS

Common Properties and Methods:

- Add() – adds an element
- Remove() – removes a specific element
- RemoveAt() – removes by index
- Count – number of elements
- Clear() – removes all elements

```
ArrayList list = new ArrayList();
list.Add(10);
list.Add(20);
```

ARRAYLIST VS ARRAY

Feature	Array	ArrayList
Size	Fixed	Dynamic
Data Type	Single type	Multiple types
Namespace	System	System.Collections
Performance	Faster	Slower (boxing/unboxing)

OPERATIONS ON ARRAYLIST

Adding Elements

```
ArrayList list = new ArrayList();
list.Add("Red");
list.Add("Blue");
```

Removing Elements

```
list.Remove("Red");    // removes value
list.RemoveAt(0);      // removes by index
```

OPERATIONS ON ARRAYLIST

Working with Range Elements

```
ArrayList list1 = new ArrayList() { 1, 2, 3 };
ArrayList list2 = new ArrayList() { 4, 5 };

list1.AddRange(list2);
```

Removing Range

```
list1.RemoveRange(1, 2); // start index, count
```

CONVERSION OF ARRAYLIST TO ARRAY

```
ArrayList list = new ArrayList() { 10, 20, 30 };
int[] array = (int[])list.ToArray(typeof(int));
```

You can convert an ArrayList into a normal array.

Useful when fixed-size or type-specific processing is required.

THANK YOU